

Quantum Gravity Wrench (QGW): Full Derivation with Mathematical Framework

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Abstract

We present the complete derivation of the Quantum Gravity Wrench (QGW), a theoretical construct designed to bridge classical general relativity and quantum field theory via non-perturbative methods. Based on a hybrid model combining Loop Quantum Gravity (LQG) spin networks and effective field theory, we map the classical concept of torque to spacetime curvature. This work includes full mathematical derivations of key equations including the torque-curvature mapping and discrete geometric operators.

1. Introduction

The pursuit of quantum gravity requires a balance between mathematical rigor and physical intuition. The QGW framework aims to achieve this by constructing spacetime from categorical principles. This paper presents the full mathematical derivation required to understand the operational mechanism.

2. Mathematical Framework

2.1 $(\infty,4)$ -Categories

We define the fundamental object C as a higher-category where: (1) Objects represent spacetime events. (2) k -morphisms ($1 \leq k \leq 4$) encode causal relations.

2.2 GH-Compactness (Locking Mechanism)

A central result is the precompactness of the morphism space $\text{Hom}_{\mathbf{C}}(X,Y)$ under the Gromov–Hausdorff metric d_{GH} , ensuring the existence of convergent subsequences.

3. Physical Action and Derivation

3.1 Discretization in LQG

In the LQG framework, QGW operation corresponds to changing the spin j on specific edges of the spin network:

$$j \rightarrow j + \delta j$$

This alters the quantum geometry, thereby affecting spacetime curvature. The area operator eigenvalues are given by:

$$\hat{A}_S \Psi = 8\pi \gamma \ell_P^2 \sqrt{j(j+1)} \Psi$$

3.2 Torque-Curvature Mapping

The fundamental operational equation linking mechanical torque τ to spacetime scalar curvature R is derived as:

$$R = (8\pi G/c^2) \cdot (\tau/V)$$

where V is the characteristic volume determined by the spin network node volume operator:

$$V_{\text{eff}} = (\gamma \ell_{\text{P}}^2)^{3/2} \sqrt{(j_{\text{eff}}^2 - j_{\text{min}}^2)}$$

3.3 Modified Einstein Field Equations

Within the QGW interaction region, the field equations are modified to include a geometric stress tensor:

$$R_{\mu\nu} - \frac{1}{2}R g_{\mu\nu} = (8\pi G/c^4) (T_{\mu\nu}^{\text{QGW}} + \tau_{\mu\nu}^{\text{geo}})$$

References

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